

DSA problems

Pattern

You are given an input string encoded using alphabet shifting, either to the right or to the left by a fixed distance. Another input will be given for the original word from the input string. Your task is to decode the input string and return the original string.

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z.

Eg.

If original string is : Testing is Boring! (This is not given)

Input will be : String s = Uftujoh jt Cpsjph! and string originalWord = "Boring";

Expected output: Testing is Boring!

(alphabet is shifted to right by 1, this difference can be anything, negative as well)

Using originalWord we have to find out the shift and return the entire original string.

Aumni & NetSpi (Exactly same question was asked in the coding round for both companies!)

Given a string, capitalize any consecutive occurrences of characters 'abeioust' that appear two or more times consecutively. Return the modified string.

For example, for the input string "Testing is boring!", the modified string is "TESTIng IS BOring".

NetSPI (interview)

Codes:

<https://www.geeksforgeeks.org/merging-intervals/>

<https://www.geeksforgeeks.org/find-the-maximum-depth-or-height-of-a-tree/>

<https://www.geeksforgeeks.org/move-zeroes-end-array/>

<https://www.geeksforgeeks.org/check-if-pair-with-given-sum-exists-in-array/>

Puzzles:

<https://www.youtube.com/watch?v=Q1B5lgxIPrU>

<https://www.careercup.com/question?id=5145121580384256>

Cumulus (Interview)

<https://www.geeksforgeeks.org/write-a-function-to-get-the-intersection-point-of-two-linked-lists/>

<https://www.geeksforgeeks.org/find-the-missing-number/>

Cerillion

Written test

Basic MCQs on java, javascript, html, aptitude, MySQL

Three queries based on joins, group by, third highest salary

Codes:

Javascript: write a function to return sum of all the elements of given array

Java: Given a string containing '"', '|' characters, and other characters, write a function to count the number of '"' characters that are not enclosed by two '|' characters.

Eg: sdf*|df*dsf|df*das*|sdf**f|

Expected output: 3

sdf*|df*dsf|df*das*|sdfdf|

Highlighted parts are enclosed between two '|'s

Java: Given a string, determine if it is a valid string, which means it contains matching pairs of parentheses, brackets, and curly braces. For example, the string 'aaa(bbb{ccc[aaa]bbb})ccc' is valid because each opening parenthesis, bracket, or curly brace has a matching closing pair. However, the string '((aaab)' is invalid because the parentheses are not correctly matched.

Interview

<https://www.geeksforgeeks.org/java-program-count-occurrences-character/>

Core Java

Latest version of java? Which one did you use in your project?

Oops concepts

Explain OOPs pillars that helped you in your project. Use cases were expected.

Abstract class, interface, multiple inheritance

Name collision when we implement two or more interfaces and they have methods with the same signatures.

Method overloading, method overriding

Upcasting, downcasting

Default methods in interface

Exception handling, try-catch-finally blocks, System.out()

What will be the output:

```
Try{
x = 1.0 / 0.0
}
Catch(RuntimeException rte)
{
sop("Runtime exception caught");
}
catch(exception e)
{
sop("Exception caught");
}
```

What will be the output:

```
System.out.println(Math.min(Double.MIN_VALUE, 0.0));
```

Threading: Why is there Runnable interface when there is already Thread class?

How is synchronisation achieved in java? Use case for synchronisation and situation where it can be inefficient.

Collection:

How do Set and HashMap work internally in Java?

What is their initial capacity and what happens when it is exhausted?

Time complexity to add an element in ArrayList, LinkedList, Set.

Is HashMap thread safe? How to make it thread safe?

Advanced java project related:

Explain the flow of a REST request in a Spring Boot application, starting from the React client to the backend DispatcherServlet, through filters (for JWT authentication/authorization), and finally to the Database.

Annotations used in the project

@Transactional use case, is it required every time? Is it method level or class level?

Where was it really required in your project, use case?

What is n+1 problem in hibernate? Did you face it in your project?

What is eager and lazy fetching? Use case for both?

What approach did you use to create tables in the database? (code-first approach with Hibernate as the ORM)

Which tools did you use for testing your project?

MySQL

Types of joins and use cases.

Stored procedures, triggers.

Consider implementing an after insert trigger on a table that adds the same entry to the same table in the trigger. What will be the result of such an implementation?

Where vs having

Group by, order by

MySQL functions, line functions, group functions examples

Drop vs delete vs truncate

DDL vs DML commands

Use of indexes? What happens internally when we create an index on a column? How to select columns to create indexes?

How do you configure a many-to-many entity association mapping?

How to rename a column?

GIT:

How would you approach making changes to the code on the main branch? Write the Git commands you would use.

SDLC:

Explain the steps of the Software Development Life Cycle (SDLC) in the context of your project, from requirement gathering to deployment.